NAME

mbpy_grdplot − Create an executable shellscript which will generate a GMT modern-mode map of gridded data in a GMT grid file or image data in a georeferenced image file.

VERSION

Version 5.0

SYNOPSIS

mbpy_grdplot **−I***grid* [**−O***root*] [**−−format***fmt*] [**−−title***text*] [**−−colorbar−label***text*] [**−−dpi***dpi*] [**−X**] [**−V**]
[**−R***xmin/xmax/ymin/ymax*] [**−J***spec*] [**−P***pagesize*] [**−−portrait** | **−−landscape**] [**−−scale−location***{l,r,t,b}*]
[**−−tick−info***spec*] [**−−length−scale***spec*]
[**−−color−mode***{1-7}*] [**−−no−color−fill**] [**−W***spec*] [**−D**] [**−Y**] [**−−stretch−color**] [**−Z***zmin/zmax*] [**−−data−scale***factor*]
[**−−magnitude***val*] [**−−azimuth***deg*] [**−−elevation***deg*] [**−−shade−flip**] [**−−intensity−file***grid*] [**−−stretch−shade**]
[**−C**[*interval*]] [**−−contour−file***grid*] [**−−contour−annotation***interval*] [**−−contour−pen***pen*] [**−−contour−gap***spec*] [**−−contour−cut***val*] [**−−contour−tick**[*spacing*]]
[**−−coast**] [**−−coast−resolution***{f,h,i,l,c}*] [**−−coast−dry−fill***color*] [**−−coast−wet−fill***color*] [**−−coast−lake−fill***color*] [**−−coast−river***color*] [**−−coast−boundary***spec*] [**−−coast−pen***pen*]
[**−−xy−overlay***file*[*:fill=..*][*:symbol=..*][*:pen=..*]] [**−−text***lon/lat/size/angle/font/justify/text*]
[**−−nav−file***datalist*] [**−−nav−format***fmt*] [**−−nav−ticks***spec*] [**−−ping−ticks***tick/annot/len*] [**−−nav−pen***pen*]
[**−−gmt−default***param/value*]

DESCRIPTION

mbpy_grdplot is a Python/PyGMT-era replacement for the Perl macro **mbm_grdplot**. It reads a grid (or, in **−−color−mode** 6/7, a georeferenced or plain image file) and generates an executable shellscript of **GMT modern mode** commands (**gmt begin** / **gmt end**, without classic mode's manual **−K**/**−O** bookkeeping) which, when run, produces a map of that data. Several styles of plots can be generated, including color fill maps, contour maps, color fill maps overlaid with contours, shaded relief color maps, slope magnitude maps, coastline overlays, swath navigation/ping-number overlays, text labels, and xy data in lines or symbols.

The plot is scaled to fit the specified page size using the same page-layout, projection-fitting, and color-palette heuristics as **mbm_grdplot**, so the two macros produce visually equivalent maps. Unlike **mbm_grdplot**'s single-letter **−M***sub***:***sub***:**... grammar for coastline/navigation/contour/xy tuning, **mbpy_grdplot** exposes each of those as its own descriptive long option (see OPTIONS below); most options are also available as classic **mbm_grdplot**-style single-letter flags where a natural equivalent exists.

mbpy_grdplot requires **GMT** 6 or later (modern mode) and a Python 3 interpreter. It depends on four sibling support modules (**mbpy_layout.py**, **mbpy_color.py**, **mbpy_misc.py**, **mbpy_grdplot_script.py**) installed alongside it; the first three are shared with the companion macros **mbpy_grd3dplot** and **mbpy_histplot**.

Unlike **mbm_grdplot**, the generated script does not include any step to display the plot on screen, and does not need a separate **gmt psconvert** step to rasterize the output -- GMT modern mode's own **gmt figure**/**gmt end** machinery renders directly to the format requested with **−−format**.

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OPTIONS

−I, **−−input** *grid*
Input grid file (or image file, for **−−color−mode** 6/7). Required.

−O, **−−output** *root*
Output root name; the generated script is *root***.sh**, the plot is *root***.***fmt*, and (unless **−W** names an existing .cpt file) the color palette is *root***.cpt**. Default: the input grid's basename with its extension removed.

−−format *fmt*
Output format passed to **gmt figure**: **pdf**, **png**, **ps**, **jpg**, **tif**, **eps**, ... Default **pdf**.

−−title *text*
Plot title, placed above the map frame.

−−colorbar−label *text*
Label for the color scale bar (the second half of **mbm_grdplot**'s **−L***title***:***label*). Default **z**.

−−dpi *dpi*
Rendering resolution: both **grdimage**'s own **−E***dpi* and the final raster's resolution (via **gmt figure**).

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print the generated script's path and a one-line summary.

−R, **−−region** *xmin/xmax/ymin/ymax*
Plot region. Default: the grid's own bounds.

−J, **−−projection** *spec*
GMT map projection/scale spec, e.g. **m1.0** (Mercator, 1 inch/degree), **x0.02** (linear, 0.02 inch/unit), **Xd6i** (linear, 6 inch width). Default: an auto-fit Mercator (geographic grids) or linear (all others) projection chosen to fill the page.

−P, **−−pagesize** *size*
Page size code: **a**−**f**, **e1** (US sizes), **a0**−**a10**, **b0**−**b10**, **c0**−**c7** (ISO/architectural), or **m1**−**m6** (large-format). Default **a**.

−−portrait, **−−landscape**
Force the page orientation instead of choosing automatically based on the plot's aspect ratio.

−−scale−location *{l,r,t,b}*
Color scale bar placement: left, right, top, or bottom. Default **b**.

−−tick−info *spec*
Raw **−B** basemap tick-interval spec, overriding the automatically computed one.

−−length−scale *spec*
A GMT distance (map) scale bar spec, passed as **−L***spec* on **gmt basemap**, e.g. **fx10/2/32.70/5.0+lkm** (**mbm_grdplot**'s **−MGL**).

−−color−mode *{1-7}*
1 plain color fill; 2 color shaded relief (synthetic illumination); 3 color shaded by a supplied intensity grid; 4 color fill of slope magnitude; 5 color shaded by slope magnitude; 6 image plot using an image's own embedded georeferencing; 7 image plot placed using **−−region** bounds (no georeferencing of its own). Default 1.

−−no−color−fill
Disable color fill entirely, for a contour-only plot.

−W, **−−color−control** *spec*
Either *style*[*/palette*[*/ncolors*]] (e.g. **1/2/11**: continuous style, palette 2, 11 colors), or the path to an existing **.cpt** file to use as-is.

−D, **−−color−flip**
Reverse the color palette.

−Y, **−−no−nice−interval**
Use a direct linear color stretch across the data range instead of a rounded ("nice") interval.

−−stretch−color
Histogram-equalize the color stretch (**−−color−mode** 1 only). With a sealevel palette (**−W***style*/**8** or **/9**), the parts of the grid below and above 0 m are equalized separately, with a color boundary exactly at 0: Haxby colors span the bathymetry bins and the land colors span the topography bins (swapped with **−D**). The color bins are shared between the two sides in proportion to their number of grid nodes. Grids with more than 16 million nodes are decimated before equalization to limit memory use.

−Z, **−−zbounds** *zmin/zmax*
Override the grid's own detected z-range for every z-dependent computation (color interval, contour interval, ...).

−−data−scale *factor*
Rescale grid values by *factor* before colorizing/shading.

−−magnitude *val***, −−azimuth** *deg***, −−elevation** *deg*
Illumination/slope-shading parameters for **−−color−mode** 2-5. Defaults depend on the mode (see **mbm_grdplot**'s **−A**).

−−shade−flip
Reverse the shading direction (negates the effective magnitude).

−−intensity−file *grid*
Intensity grid for **−−color−mode** 3.

−−stretch−shade
Histogram-equalize the intensity file first (**−−color−mode** 3 only).

−C, **−−contour** *[interval]*
Enable a **grdcontour** overlay, at the given interval/spec/**.cpt** path if supplied, otherwise at **mbm_grdplot**'s own auto-computed "reasonable" interval for the data range.

−−contour−file *grid*
Grid to contour, if different from **−−input**.

−−contour−annotation *interval*
Contour annotation interval (**mbm_grdplot**'s **−MCA**).

−−contour−pen *pen*
Contour line pen (**−MCW**). Default **c1p** (the regular, i.e. non-annotated, contour pen at 1 point).

−−contour−gap *spec*
Contour line gap spec (**−MCG**).

−−contour−cut *val*
Omit contour segments shorter than *val* (**−MCQ**).

−−contour−tick *[spacing]*
Add downhill-direction ticks to closed contours, optionally at the given tick spacing (**−MCT**).

−−coast
Enable a coastline overlay (**gmt coast**).

−−coast−resolution *{f,h,i,l,c}*
Coastline database resolution: full, high, intermediate, low, crude.

−−coast−dry−fill, **−−coast−wet−fill**, **−−coast−lake−fill** *color*
Fill colors for land, water, and lakes.

−−coast−river *color*
Draw rivers in the given color.

−−coast−boundary *spec*
Draw a national boundary; repeatable.

−−coast−pen *pen*
Pen for the coastline itself.

−−xy−overlay *file***[:fill=***fill***][:symbol=***symbol***][:pen=***pen***]**
Overlay an xy data file (**gmt plot**), with optional colon-separated styling, e.g. **points.xy:fill=red:symbol=c0.1:pen=1p,blue**. Repeatable.

−−text *lon/lat/size/angle/font/justify/text*
Place a text label (**gmt text**), e.g. **−122.2/36.6/12/0/1/LB/Monterey Bay**. Repeatable.

−−nav−file *datalist*
MB-System swath datalist to plot navigation from, via MB-System's own **gmt mbcontour** GMT supplement module.

−−nav−format *fmt*
Datalist format code. Default: auto-detected via **mbformat**.

−−nav−ticks *spec*
Tick/annotation spec, e.g. **0.25/1/4/0.15**, or **F**/**FP** to annotate with the swath file name (perpendicular, for **FP**).

−−ping−ticks *tick/annot/len*
Ping-number annotation spec along the navigation track, e.g. **50/100/0.1**.

−−nav−pen *pen*
Pen for the navigation track line.

−−gmt−default *param/value*
Override a GMT default parameter for this plot (e.g. **MAP_FRAME_TYPE/plain**); repeatable.

EXAMPLES

Suppose one wants to generate a shaded-by-slope-magnitude topography map with a swath navigation overlay, a distance scale bar, and a color scale bar labeled in meters, rendered directly to a 300 dpi PNG:

```
mbpy_grdplot −I topo.grd −O topoplot \
  −−color−mode 5 −−shade−flip −−magnitude 1 \
  −−title "Survey Area" −−colorbar−label "Topography (meters)" \
  −−length−scale "fx10/2/32.70/5.0+lkm" \
  −−nav−file datalist.mb−1 \
  −−pagesize c −−dpi 300 −−format png −−execute −V
```

This produces **topoplot.sh** (the generated GMT modern-mode shellscript), **topoplot.cpt** (the color palette), and, because **−−execute** was given, runs the script immediately to produce **topoplot.png**.

A plain color-fill map with a 100-meter contour overlay, written to a script for later inspection/execution rather

than run immediately:

```
mbpy_grdplot −I topo.grd −O topocont \
    −−color−mode 1 −−contour 100 −−contour−pen "0.5p,black" \
    −−title "Survey Area" −−colorbar−label "Topography (meters)" \
    −−format pdf −V
```

SEE ALSO

mbm_grdplot(1), **mbpy_grd3dplot**(1), **mbpy_histplot**(1), **mbpy_xyplot**(1), **grdimage**(1), **grdcontour**(1), **mbdefaults**(1), **mbgrid**(1), **mbsystem**(1), **mbformat**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).